AXI2AHB-Lite Bridge Specification & Test Report
Document Version: V1.0
Date: 2026.03.26
Target Design: Parameterized AXI Slave to AHB-Lite Master Bridge

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1. Overview

This design implements a **parameterized AXI-to-AHB-Lite protocol bridge** that receives AXI read/write requests, buffers transactions, converts AXI bursts into a sequence of AHB-Lite beats, and returns AXI-compliant B/R responses. It strictly follows the AHB-Lite single-transaction execution model and maintains AXI protocol compliance.

- Top-level file: dut/axi2ahb_bridge/axi2ahb_bridge_top.sv
- Core function: AXI burst → AHB-Lite beat conversion & protocol bridging

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2. Module Architecture

The bridge consists of **four major building blocks** with clear partitioning and dedicated file paths.

Block Name	Key File Path	Core Function
AXI Frontend	axi_frontend.sv	Capture AXI requests; generate AXI B/R responses
Request/Response FIFO	fifo_wrapper.sv	Independent synchronous FIFOs for AW/W/AR/R; depth configurable
Bridge Control Core	bridge_core.sv	Request arbitration; burst-to-beat expansion; execution control
AHB Backend	ahb_backend.sv	AHB-Lite beat execution; drive AHB bus signals

2.1 AXI Frontend Sub-functions

1. **axi_req_capture**: Capture AW/W/AR transactions, perform legality check, push to FIFOs
2. **axi_rresp_gen**: Drive AXI R channel directly from read-response FIFO
3. **axi_bresp_gen**: Single-entry write response buffer

2.2 Bridge Control Core Sub-functions

1. **bridge_fifo_ctrl**: Evaluate request validity & resource availability
2. **bridge_arbiter**: Fair arbitration between read/write requests
3. **bridge_controller**: Expand AXI bursts to AHB beats; track execution

2.3 AHB Backend

- Core executor: ahb_beat_executor.sv
- Execution mode: One AHB-Lite beat at a time
- Driven signals: HADDR, HTRANS, HWRITE, HSIZE, HBURST, HWDATA, HSTRB

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3. Functional Behavior

3.1 Supported Features

- AXI burst types: **FIXED, INCR, WRAP**
- Transfer size: All aligned sizes within data bus width

- Narrow transfers: Supported (aligned sub-word writes on wide buses)
- AXI channels: Full AW/W/B/AR/R implementation

3.2 Unsupported Features

- **Unaligned transfers**: Start address unaligned to beat size → illegal request
- AXI ID tracking / reordering / interleaving: Not implemented

3.3 Illegal Request Handling

Transaction Type	Handling Logic
Illegal Read	No AHB traffic; return **SLVERR** directly
Illegal Write	Drain W beats; no AHB execution; return **SLVERR**

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4. Ordering & Throughput Model

1. **Request buffering**: Configurable FIFO depth for queuing incoming requests
2. **Execution model**: **Strictly serialized** (only one active transaction at a time)
3. **Response ordering**: In-order completion; natural AXI response ordering
4. **Response buffer**:
    - Write response: Single-entry buffer
    - Read response: FIFO-buffered, FWFT direct drive

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5. Protocol & Feature Limits

- Parameterization: Address width, data width, strobe width, FIFO depths
- AHB HBURST: Fixed to **INCR**; AXI burst semantics preserved by controller address logic
- Write capture: New AW accepted only when prior write burst capture completes

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6. Concise Specification Summary

✅ AXI Slave → AHB-Lite Master bridge
✅ FIXED/INCR/WRAP bursts supported
✅ Aligned transfers & narrow transfers supported
❌ Unaligned transfers prohibited
✅ Configurable FIFO for AW/W/AR/R
✅ Serial execution matching AHB-Lite
❌ No AXI ID / out-of-order support
✅ Full exception handling (SLVERR for illegal requests)

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7. Test Plan & Verification Result

7.1 Test List & Coverage

All tests passed; full functional & exception coverage achieved.

Test Case File	Test Objective	Test Result
base_test.sv	Basic environment & connectivity check	Pass
single_write_read_test.sv	Basic single write & read function	Pass

bd_write_fd_read_test.sv	Backpressure write + forward read	Pass
fd_write_bd_read_test.sv	Forward write + backpressure read	Pass
incr_random_len_size_wr_test.sv	INCR burst random length/size write	Pass
wrap_random_len_size_wr_test.sv	WRAP burst random length/size write	Pass
fixed_random_len_size_wr_test.sv	FIXED burst random length/size write	Pass
random_traffic_test.sv	Random AXI traffic test	Pass
mixed_random_traffic_test.sv	Mixed read/write random traffic	Pass
frontend_exception_test.sv	Frontend illegal request handling	Pass
backend_exception_test.sv	Backend exception & error response	Pass
fifo_full_stress_test.sv	FIFO full backpressure stress test	Pass
reset_recovery_test.sv	Reset sequence & recovery function	Pass

7.2 Test Conclusion

1. **Function**: AXI-to-AHB-Lite conversion, burst splitting, response generation fully correct
2. **Protocol**: AXI/AHB-Lite signal behavior compliant; no protocol violation
3. **Exception**: Illegal requests, FIFO full, reset handled as spec
4. **Reliability**: Stress & recovery tests passed; stable under corner conditions

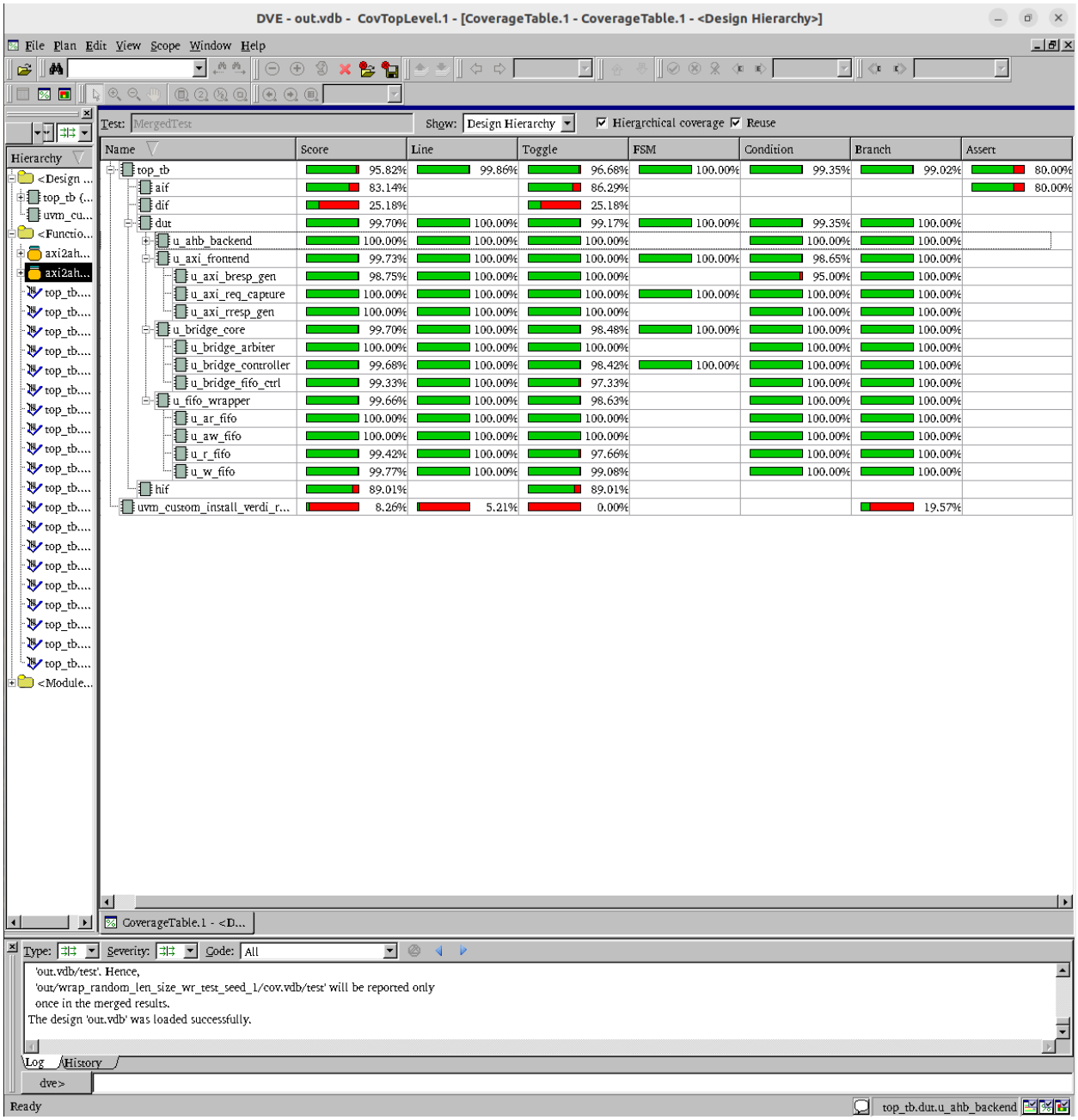

Figure1.Code coverage

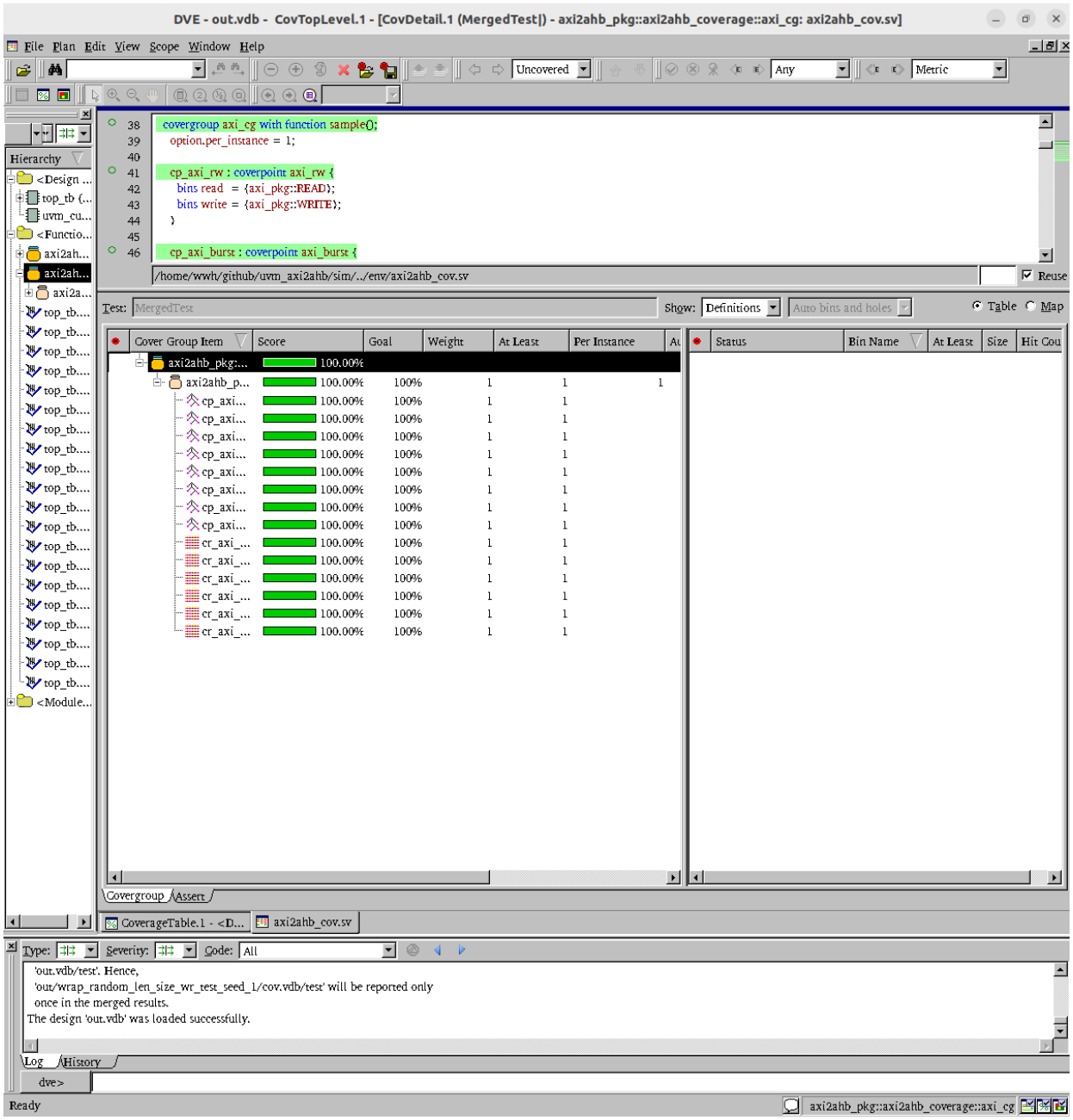

Figure2. Functional coverage